Short-Term Traffic Flow Forecasting with XGBoost and Traffic Condition Classification via Gradient Boosting Machines

Duy-Anh Nguyen[1, *[0000-0002-2432-2050]], Minh-Nhat That[1], Tat-Hien Le[1[0000-0001-8616-3254]], Thanh-Nhat Luong[1[0009-0005-7097-1739]], and Nguyen Duy Anh[1, **[0000-0002-5280-8453]]

[1] Ho Chi Minh City University of Technology (HCMUT),
Viet Nam National University Ho Chi Minh City, Vietnam
*ndanhcdt@hcmut.edu.vn,**duyanhnguyen@hcmut.edu.vn

Abstract. This paper presents a short-term traffic flow prediction and traffic situation classification approach using XGBoost and Gradient Boosting Machines (GBM). The proposed methodology leverages historical traffic flow data and time-related features to accurately forecast traffic flow and classify traffic conditions. Experiments on a real-world dataset and simulations in the SUMO environment demonstrate the effectiveness of the approach. The method shows promise for enhancing intelligent transportation systems and aiding in proactive traffic management.

Keywords: Traffic flow prediction, XGBoost, Gradient Boosting Machines, Traffic situation classification.

1 Introduction

Efficient traffic management and accurate traffic flow prediction are critical for alleviating congestion, reducing travel times, and enhancing overall transportation efficiency in urban areas. Short-term traffic flow prediction, which aims to forecast traffic conditions in the near future (e.g., next 5-30 minutes), plays a vital role in proactive traffic management and intelligent transportation systems [1]. Accurate predictions enable traffic controllers to make informed decisions, optimize signal timings, and provide real-time information to road users [2].

Traditional traffic flow prediction methods, such as time series models and statistical approaches, often struggle to capture the complex and dynamic nature of traffic patterns [3]. In recent years, machine learning techniques have emerged as powerful tools for traffic flow prediction, owing to their ability to learn intricate patterns and dependencies

from historical traffic data [4]. Among various machine learning algorithms, ensemble methods, such as Gradient Boosting Machines (GBM) and Extreme Gradient Boosting (XGBoost), have shown promising results in handling complex non-linear relationships and achieving high prediction accuracy [5].

This paper presents a novel approach for short-term traffic flow prediction that combines the strengths of XGBoost for prediction and GBM for traffic situation classification. The proposed method aims to leverage the capabilities of these ensemble learning techniques to capture the underlying patterns in traffic data and provide accurate forecasts. By incorporating traffic situation classification using GBM, the model can better understand the contextual information and adapt to different traffic conditions.

The main contributions of this paper are threefold. First, we propose a hybrid approach that integrates XGBoost for traffic flow prediction and GBM for traffic situation classification, enabling the model to capture both temporal dependencies and contextual information. Second, we evaluate the performance of the proposed method on a public dataset containing traffic flow information collected over a period of 2 months. Third, we simulate the proposed model in the SUMO (Simulation of Urban Mobility) traffic simulation environment to demonstrate its effectiveness for short-term traffic flow prediction in a realistic scenario.

SUMO is an open-source, microscopic traffic simulation package that allows for the modeling and simulation of real-world traffic scenarios. It provides a flexible and extensible framework for evaluating traffic management strategies, testing intelligent transportation systems, and validating traffic prediction models. By simulating the proposed traffic flow prediction model in SUMO, we can assess its performance in a dynamic and realistic traffic environment, considering various factors such as road network topology, traffic demand, and vehicle behavior.

The remainder of this paper is organized as follows. Section 2 reviews related work on traffic flow prediction and ensemble learning methods. Section 3 describes the proposed methodology, including problem formulation, data preprocessing, XGBoost-based traffic flow prediction, GBM-based traffic situation classification, and model evaluation metrics. Section 4 presents the experimental results and discussion, including dataset description, performance evaluation, comparative analysis, and insights gained. Finally, Section 5 concludes the paper and outlines future research directions.

2 Related Work

2.1 Traffic Flow Prediction Models

Traffic flow prediction has been an active area of research due to its importance in intelligent transportation systems. Various approaches have been proposed to tackle this problem, ranging from classical statistical methods to advanced machine learning techniques.

Time series models, such as Autoregressive Integrated Moving Average (ARIMA) and its variants, have been widely used for traffic flow prediction [6]. These models capture the temporal dependencies in traffic data and provide short-term forecasts based on historical observations. However, they often struggle to capture complex non-linear relationships and are sensitive to sudden changes in traffic patterns [7].

Machine learning methods have gained popularity in recent years for traffic flow prediction. Artificial Neural Networks (ANNs) have been applied to learn the intricate patterns and dependencies in traffic data [8]. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, have shown promising results in capturing long-term dependencies and handling sequential data [9].

Deep learning models, such as Convolutional Neural Networks (CNNs) and Graph Convolutional Networks (GCNs), have also been explored for traffic flow prediction [10]. These models can capture spatial dependencies and learn hierarchical features from traffic data. Hybrid models combining different deep learning architectures have been proposed to leverage the strengths of multiple models [11].

2.2 Ensemble Learning Methods

Ensemble learning methods have gained attention in various domains, including traffic flow prediction, due to their ability to combine multiple base learners and improve prediction accuracy. Gradient Boosting Machines (GBM) is a popular ensemble learning technique that sequentially constructs a set of weak learners, typically decision trees, to minimize a loss function [12]. GBM has been successfully applied in traffic flow prediction, demonstrating its effectiveness in capturing complex patterns and handling non-linear relationships [13]. Extreme Gradient Boosting (XGBoost) is an optimized implementation of gradient boosting that has gained widespread popularity due to its exceptional performance and scalability [14]. XGBoost incorporates regularization techniques and a second-order approximation of the objective function to prevent overfitting and improve generalization ability. It has been applied in various domains, including traffic flow prediction, and has shown state-of-the-art results [15].

Other ensemble learning methods, such as Random Forests and AdaBoost, have also been explored for traffic flow prediction [16]. These methods combine multiple decision trees or weak learners to make robust predictions and handle the inherent variability in traffic data. Despite the advancements in traffic flow prediction models and ensemble learning methods, there is still room for improvement in terms of prediction accuracy, robustness, and adaptability to different traffic conditions. This paper aims to address these challenges by proposing a hybrid approach that combines XGBoost for prediction and GBM for traffic situation classification, leveraging the strengths of both techniques to enhance short-term traffic flow prediction performance.

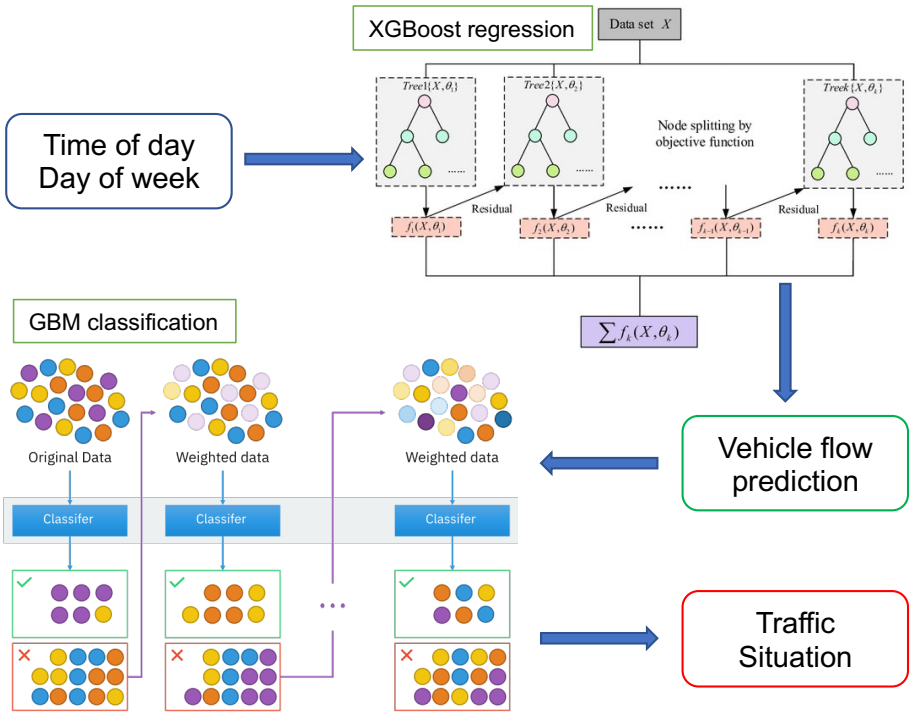


Fig. 1. Ensemble Learning Process with XGBoost for Regression and GBM for classification

3 Methodology

Figure 1 provides an overview of the proposed methodology for short-term traffic flow prediction using XGBoost regression and traffic situation classification using Gradient Boosting Machines (GBM). The diagram shows the flow of data and the key components of the system. The input data consists of traffic flow information collected from sensors or detectors, which is then preprocessed and transformed into a suitable format for the models. The preprocessed data is fed into two separate branches: XGBoost regression for traffic flow prediction and GBM for traffic situation classification. In the XGBoost regression branch, the historical traffic flow data, along with relevant features like time of day and day of the week, is used to train the XGBoost model. The model learns to predict the traffic flow values for different vehicle types (car, bike, bus, truck) and the total traffic flow for the next time steps (e.g., 15, 30, and 60 minutes ahead). In the GBM branch, the traffic flow data, along with contextual features, is used to train the GBM model for traffic situation classification. The GBM model learns to classify the traffic situation into predefined categories such as low, normal, high, or heavy.

3.1 Dataset Description

The dataset used in this study contains traffic flow information collected at a specific junction over a period of 2 months from Kaggle [17]. The data is recorded at a 15-minute sampling interval, resulting in 96 data points per day (4 data points per hour). The dataset includes the following features:

- Time: The timestamp of the recorded data in the format of "HH:MM:SS AM/PM".
- Date: The date of the recorded data in the format of "DD" (day of the month).
- Day of the week: The corresponding day of the week for each recorded data point (e.g., Monday, Tuesday, etc.).
- CarCount: The number of cars counted at the junction.
- BikeCount: The number of bikes counted at the junction.
- BusCount: The number of buses counted at the junction.
- TruckCount: The number of trucks counted at the junction.
- Total: The total number of vehicles (sum of cars, bikes, buses, and trucks) counted at the junction.
- Traffic Situation: The traffic situation at the junction, labeled as "low", "normal", "high" or "heavy".

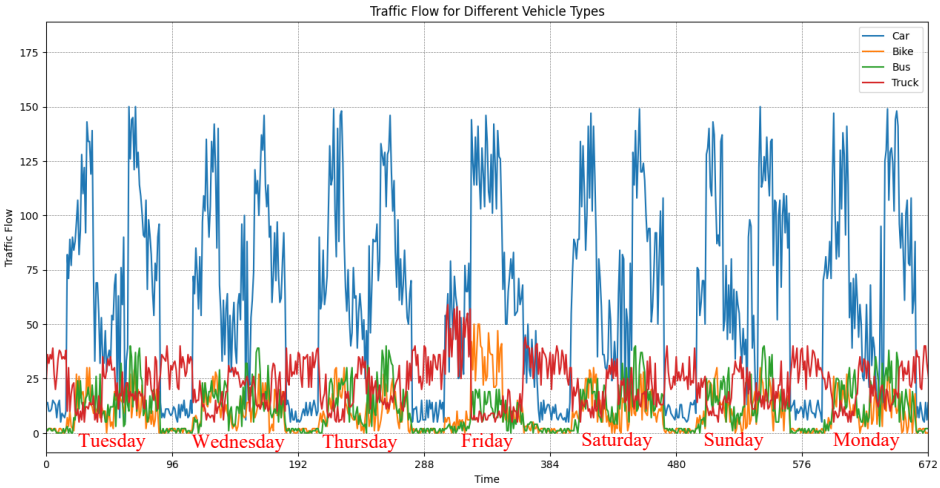


Fig. 2. The number of different type of vehicles during a week from Kaggle dataset.

The dataset captures detailed traffic patterns at a junction, with time-stamped vehicle counts allowing analysis of daily and weekly trends and the impact on traffic conditions. The 15-minute intervals are ideal for studying short-term dynamics and building traffic prediction and optimization models.

Figure 2 depicts the traffic flow patterns for different vehicle types (car, bike, bus, truck) over a week fromt he dataset. Distinct peaks and troughs can be observed, corresponding to the typical times when people commute to and from work. The car traffic flow (blue line) has the highest volume and clearly shows the daily patterns. On

weekdays, there are two prominent peaks in car traffic flow. The first peak occurs around 7:30 AM to 8:45 AM, attributed to people commuting to work or school during the morning rush hour. The second peak is observed around 5:30 PM to 6:15 PM, likely representing people returning home from work during the evening rush hour. On weekends (Saturday and Sunday), the traffic flow patterns for cars differ, with the morning peak occurring slightly later (8:45 AM to 10:00 AM) and being less pronounced, suggesting that people tend to start their activities later compared to weekdays. The evening peak on weekends is also less distinct and more spread out, indicating a more varied pattern of activities and travel times. Other vehicle types (bikes, buses, trucks) show similar daily variations, although with lower volumes compared to cars. Truck traffic flow appears to have a more consistent pattern throughout the day, likely due to commercial and logistics activities.

Traffic peak times and intensity can differ by area, affected by infrastructure and local factors. Yet, most urban regions experience morning and evening rush hours tied to work commutes on weekdays. Analyzing these patterns is vital for effective traffic management, optimizing signals, reducing congestion, and enhancing public transit for high-demand times.

3.2 Problem Formulation

Consider a single road junction where traffic flow data is collected over a period of 2 months, with a 15-minute sampling interval. The dataset includes features such as time of day, day of the week, vehicle counts for different types (car, bike, bus, truck), and total vehicle count. The traffic situation at the junction is also labeled as low, normal, high, or heavy for each 15-minute interval.

The objective of the short-term traffic flow prediction problem is to forecast the traffic flow values (CarCount, BikeCount, BusCount, TruckCount) for the next H time steps, given the historical traffic flow data and other relevant features such as time of day, day of the week. Formally, the goal is to learn a mapping function $f(\cdot)$ that takes the historical traffic flow data $X^{(t-T+1):t}$ and other relevant features F^t as input and predicts the future traffic flow values $X^{(t+1):(t+H)}$ as output, where T is the length of the historical time window and H is the prediction horizon:

$$X^{(t+1):(t+H)} = f(X^{b(t-T+1):t}, F^t) \tag{1}$$

Additionally, the traffic situation classification problem aims to classify the traffic situation S^t at each time step t into different categories (e.g., low, normal, high, heavy) based on the predicted vehicle count from the prediction model above. The objective is to learn a mapping function $g(\cdot)$ that takes the predicted traffic flow data $X^{(t+1):(t+H)}$ and other relevant features F^t as input and predicts the traffic situation S^t as output:

$$S^{(t+1):(t+H)} = g(X^{(t+1):(t+H)}, F^t) \tag{2}$$

The proposed approach addresses both the traffic flow prediction and traffic situation classification problems simultaneously. By incorporating the predicted traffic situation information into the prediction model, we aim to improve the accuracy and adaptability of the traffic flow forecasts.

The proposed approach addresses both the traffic flow prediction and traffic situation classification problems simultaneously. By incorporating the predicted traffic flow as an additional input feature for the traffic situation classification model, we aim to improve the accuracy and adaptability of the forecasts.

The objective is to minimize the prediction error between the predicted traffic flow values and the actual traffic flow values, as well as maximize the classification accuracy of the traffic situation. The prediction error can be measured using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), while the classification accuracy can be evaluated using metrics such as precision, recall, and F1-score.

3.3 Data Preprocessing

The data preprocessing steps for the traffic flow dataset include time transformation, day of week encoding, and data splitting. The 'Time' column is converted from the "HH:MM:SS AM/PM" format to the number of seconds from midnight, allowing for easier manipulation and analysis of time-related features. The 'Day of Week' column is encoded using sine and cosine transformations to capture the cyclical nature of the days, creating two new features, *'sin_day'* and *'cos_day'*, which represent the sine and cosine values of the day of the week, respectively. This encoding helps the models better capture the cyclical patterns and relationships associated with different days of the week. Finally, the preprocessed dataset is split into training and testing sets, with a common split ratio of 80% for training and 20% for testing, which can be adjusted based on the dataset size and specific requirements. The training set is used to train the GBM and XGBoost models, while the testing set is used to evaluate their performance, ensuring that the testing set remains unseen during the model training process to assess the model's generalization ability.

3.4 Short-Term Traffic Flow Prediction using XGBoost

XGBoost, an optimized gradient boosting algorithm, is employed for short-term traffic flow prediction. The XGBoost model is trained to forecast the traffic flow values for each vehicle type (CarCount, BikeCount, BusCount, TruckCount) for the next H time steps, based on historical traffic flow data and time-related features. The input features for the XGBoost model include lagged traffic flow values and time-related features (e.g., seconds from midnight, sine and cosine transformations of the day of the week). The model learns the complex non-linear relationships between the input features and the target traffic flow values, capturing the temporal dependencies. The XGBoost model is trained using the training set and hyperparameter tuning techniques, such as grid search or random search, to optimize its performance. The trained model is then evaluated on the testing set using appropriate evaluation metrics, such as Mean Absolute Error (MAE), Mean Squared Error (MSE), or Root Mean Squared Error (RMSE), to assess its prediction accuracy and generalization ability. The predicted traffic flow values from the XGBoost model are then used as input features for the GBM model, along with other relevant features, to classify the traffic situation at each time step.

3.5 Traffic Situation Classification using GBM

Gradient Boosting Machines (GBM) is employed for traffic situation classification. The GBM model is trained to predict the traffic situation (e.g., low, normal, high, or heavy) at each time step based on the predicted traffic flow values obtained from the XGBoost model and other relevant features such as time of day and day of the week. The input features for the GBM model include the predicted traffic flow values for each vehicle type from the XGBoost model, as well as time-related features (e.g., seconds from midnight, sine and cosine transformations of the day of the week). The GBM model learns the relationship between the input features and the corresponding traffic situation labels, capturing the patterns and dependencies that indicate different traffic conditions. The model is trained using the training set, and hyperparameter tuning techniques, such as grid search or random search, are applied to optimize its performance. The trained GBM model is then evaluated on the testing set using appropriate evaluation metrics, such as accuracy, precision, recall, and F1-score, to assess its classification performance. The predicted traffic situations from the GBM model provide valuable insights into the overall traffic conditions at each time step, complementing the traffic flow predictions from the XGBoost model.

3.6 SUMO Traffic Simulation

To evaluate the effectiveness of the proposed short-term traffic flow prediction and traffic situation classification models in a realistic setting, we utilize the SUMO (Simulation of Urban Mobility) traffic simulation environment. The simulation scenario consists of a single junction where two roads intersect, with each road having two lanes per direction. The junction accommodates four types of vehicles: cars, bikes, buses, and trucks. The simulation is set up to replicate the traffic flow patterns observed in the dataset (figure 3). The number of vehicles of each type entering the junction is determined based on the corresponding traffic flow values at each time step in the dataset. The simulation progresses in 900-second (15-minute) intervals, aligning with the time resolution of the dataset.

At each time step, the trained XGBoost model is applied to predict the traffic flow values for each vehicle type for the next 15 minutes, 30 minutes, 1 hour, and 2 hours. These predictions are based on the historical traffic flow data and time-related features used to train the model. Simultaneously, the trained GBM model is employed to classify the traffic situation at each time step, considering the predicted traffic flow values from the XGBoost model and other relevant features such as time of day and day of the week. The GBM model predicts the traffic situation providing insights into the overall traffic conditions at the junction.

The SUMO simulation also provides valuable insights into the impact of the predicted traffic flow and classified traffic situations on the overall traffic dynamics at the junction. It enables the evaluation of various performance metrics, such as average vehicle waiting time, queue lengths, and throughput, under different traffic scenarios. This

information can assist traffic management authorities in making informed decisions regarding traffic control strategies, signal timing optimization, and resource allocation.

Furthermore, the SUMO simulation environment allows for the testing and validation of the proposed models under varying traffic demand levels, road network configurations, and vehicle behaviors. By modifying the simulation parameters, we can assess the robustness and adaptability of the models to different traffic conditions and identify potential areas for improvement.

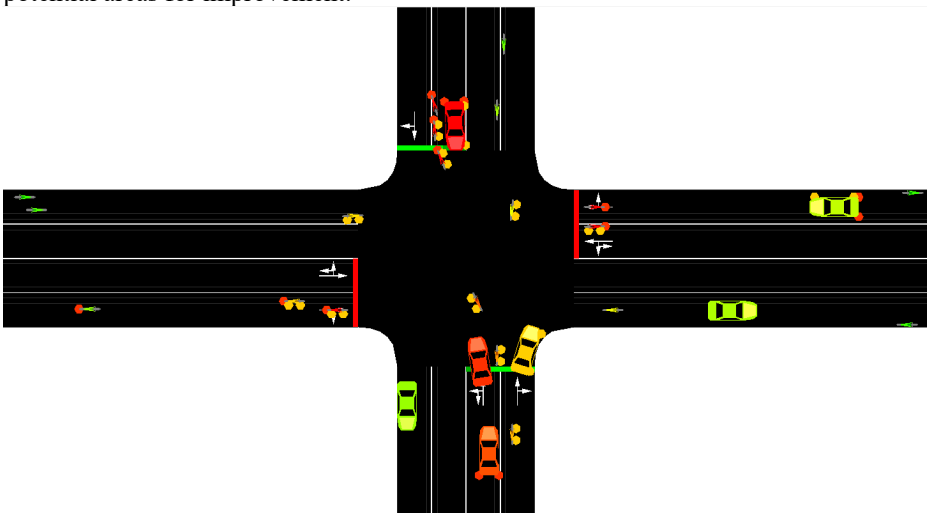


Fig. 3. SUMO simulation scenario consists of a single junction.

4 Experimental Results and Discussion

4.1 Traffic flow prediction

To evaluate the performance of various models for traffic flow prediction, we conducted experiments on the real-world traffic dataset. The models considered in this study include Linear Regression, Decision Tree, Random Forest, Gradient Boosting, AdaBoost, SVR, KNN, and XGBoost. Table 5 presents the comparative results of these models (for CarCount) in terms of Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared (R2) metrics. The XGBoost model achieves the lowest MSE of 412.7004 and MAE of 14.8182, indicating its superior performance in accurately predicting traffic flow compared to other models. XGBoost also demonstrates a high R2 value of 0.7917, suggesting a strong fit to the data and its ability to explain a significant portion of the variance in the traffic flow. Other ensemble models, such as AdaBoost, Gradient Boosting, and Extra Trees, also show competitive performance with relatively low MSE and MAE values. SVR exhibits the highest MSE and MAE among the compared models, indicating its limited effectiveness in capturing the complex patterns and dependencies in the traffic flow data. Linear regression-based models provide a baseline performance but are outperformed by the ensemble models. Decision Tree and

Random Forest, despite their simplicity, achieve reasonable results and can be considered as viable alternatives when interpretability is a priority.

Table 1. Table captions should be placed above the tables.

No.	Model	MSE	MAE	R2
1	Linear Regression	1941.08	37.85	0.02
2	Decision Tree	420.23	14.96	0.79
3	Random Forest	422.01	14.99	0.79
4	Gradient Boosting	432.85	15.61	0.78
5	AdaBoost	933.47	23.36	0.53
6	SVR	1418.88	31.10	0.28
7	KNN	424.42	14.97	0.79
8	XGBoost	412.70	14.82	0.79

The experimental results highlight the superiority of the XGBoost model in accurately predicting traffic flow, making it a suitable choice for the proposed approach. The combination of XGBoost's ability to handle complex non-linear relationships and its robustness to outliers and noise contributes to its outstanding performance in traffic flow prediction tasks.

Figure 4 illustrates the predicted traffic flow versus the actual traffic flow for a specific day in the test set using the XGBoost model. The predicted flow closely follows the actual flow, capturing the temporal patterns and variations throughout the day.

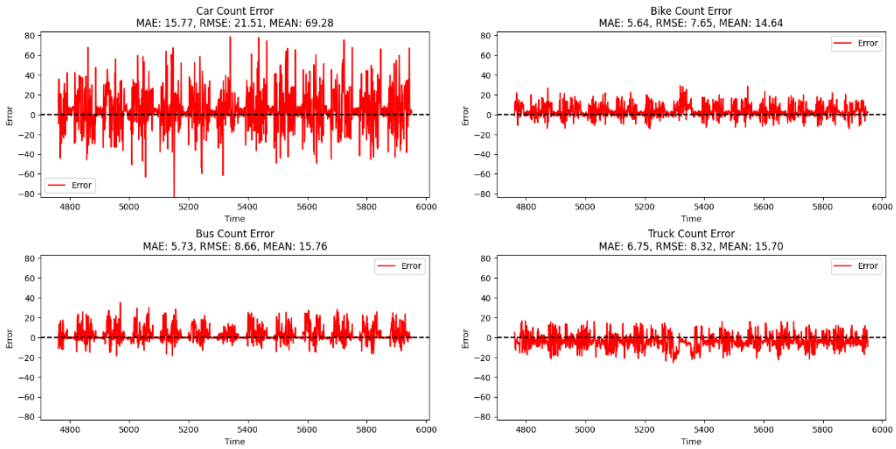


Fig. 4. The discrepancy in vehicle counts as predicted by the model versus those observed in the test set.

4.2 Traffic Classification Performance

The effectiveness of various machine learning models for traffic situation classification was evaluated using same dataset. Table 2 presents the test accuracy of different models, including K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Decision Tree, Random Forest, Gradient Boosting Machines (GBM), and XGBoost. Among the compared models, GBM achieves the highest test accuracy of 0.972, outperforming other models by a significant margin. XGBoost also demonstrates strong performance, closely following GBM, Decision Tree and Random Forest. In contrast, KNN and SVM show lower accuracy, indicating their limited effectiveness in classifying traffic situations based on the given features.

Table 2. Test accuracy of various machine learning models for traffic situation classification.

No.	Model	Test Accuracy
1	KNN	0.861
2	SVM	0.628
3	Decision Tree	0.919
4	Random Forest	0.906
5	Gradient Boosting Machines	0.972
6	XGBoost	0.969

The superior performance of Gradient Boosting Machines (GBM) and XGBoost is due to their ensemble methods, which build models that learn from and correct the errors of previous ones. This is particularly effective for traffic datasets, which often contain complex and non-linear patterns. Decision Tree and Random Forest, being tree-based models, are effective in handling non-linear relationships but may be prone to overfitting. In contrast, KNN struggles with high-dimensional data and SVM with overlapping categories, which are common in traffic situations, leading to their lower accuracy in this context.

Table 3. Classification report of Gradient Boosting Machines.

Label	Precision	Recall	F1-score	Support
Low	0.99	0.97	0.98	146
Normal	0.98	0.99	0.98	748
High	0.91	0.84	0.87	81
Heavy	0.97	0.96	0.96	216
Accuracy			0.97	1191
Macro avg	0.96	0.94	0.95	1191
Weighted avg	0.97	0.97	0.97	1191

The classification report for Gradient Boosting Machines (GBM) indicates high effectiveness in traffic situation classification, with outstanding precision and recall for 'Low' and 'Normal' traffic conditions, as evidenced by F1-scores of 0.98 for both. The model slightly underperforms in 'High' traffic situations with an F1-score of 0.87, which still denotes a good predictive ability. 'Heavy' traffic classification is also strong, showing near-perfect precision and recall. Overall, GBM achieves a 97% accuracy, with a weighted average F1-score of 0.97, suggesting that the model is particularly adept at handling classes with more instances ('Normal' and 'Heavy') and may benefit from additional data or tuning for classes with fewer instances ('High'). The **Macro average** scores, which calculate the metric independently for each class and then take the average, treating all classes equally, show slightly lower values than the weighted average, indicating that while the model performs exceptionally well overall, there may be a small performance drop when considering all classes equally, possibly due to class imbalance.

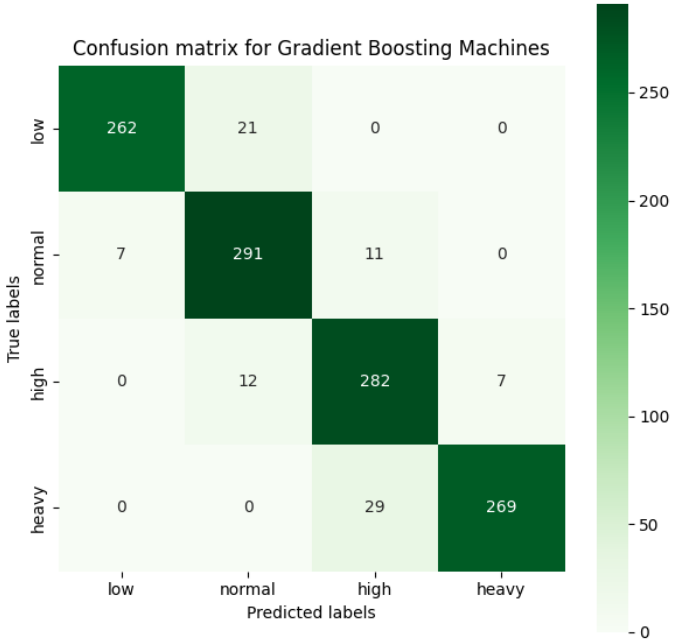


Fig. 5. Confusion matrix for the Gradient Boosting Machines (GBM) model in traffic situation classification

4.3 SUMO Traffic Simulation Results

The SUMO traffic simulation results demonstrate the effectiveness of the proposed traffic flow prediction and traffic situation classification models. The models were evaluated at different time horizons, achieving accuracy rates of 77.89%, 77.66%, 78.26%, and 77.27% for 15-minute, 30-minute, 1-hour, and 2-hour predictions, respectively (Table 4). These high accuracy rates validate the models' ability to capture traffic

patterns and dynamics accurately. The simulation environment allows for the evaluation of the models' performance under various traffic scenarios, enabling the assessment of their robustness and adaptability. The insights gained from the simulation results can assist traffic management authorities in making informed decisions for traffic control strategies and resource allocation. However, further validation using diverse datasets and real-world implementation would be beneficial to assess the generalizability and practical applicability of the proposed models. Overall, the SUMO traffic simulation results highlight the potential of the proposed models in contributing to the development of more efficient and sustainable transportation networks.

Table 4. Accuracy of traffic situation predictions at multiple time horizons.

Prediction Horizon	Total Predictions	Correct Predictions
15 minutes	95	74
30 minutes	94	73
1 hour	92	72
2 hours	88	68

5 Conclusion and Future Work

The experimental results and SUMO traffic simulation demonstrate the effectiveness of the proposed approach for short-term traffic flow prediction and traffic situation classification. The XGBoost model achieves high accuracy in predicting traffic flow for different vehicle types and total traffic flow, outperforming other baseline models. The GBM model exhibits strong performance in classifying traffic situations, with high precision, recall, and F1-score values across all traffic situation labels.

The superiority of XGBoost and GBM can be attributed to their ability to capture complex non-linear relationships, handle high-dimensional data, and leverage the collective knowledge of multiple weak learners. The ensemble nature of these models enables them to learn intricate patterns and dependencies in the traffic data, resulting in accurate predictions and classifications.

However, there are limitations and areas for future improvement. The current approach does not consider spatial dependencies and correlations between neighboring road segments, which could enhance the models' performance. Incorporating real-time data from multiple sources and developing adaptive models to handle dynamic changes in the road network would make the system more robust and applicable to real-time traffic management.

Despite these limitations, the proposed XGBoost-GBM approach demonstrates promising results for short-term traffic flow prediction and traffic situation classification. The findings contribute to the development of intelligent transportation systems and can assist traffic authorities in making informed decisions for traffic control and management.

Future work should focus on incorporating long-term traffic patterns, integrating real-time data from multiple sources, considering spatial dependencies, and enhancing the interpretability of the models. Additionally, the scalability and real-time performance of the proposed approach should be investigated to ensure its applicability in large-scale traffic networks and real-time traffic management systems.

Acknowledgements

We acknowledge the support of time and facilities from Ho Chi Minh City University of Technology (HCMUT), Viet Nam National University Ho Chi Minh City (VNU-HCM) for this study.

References

1. Smith, B.L., Williams, B.M., & Oswald, R.K. (2002). Comparison of parametric and nonparametric models for traffic flow forecasting. Transportation Research Part C: Emerging Technologies, 10(4), 303-321.
2. Vlahogianni, E.I., Karlaftis, M.G., & Golias, J.C. (2014). Short-term traffic forecasting: Where we are and where we're going. Transportation Research Part C: Emerging Technologies, 43, 3-19.
3. Lana, I., Del Ser, J., Velez, M., & Vlahogianni, E.I. (2018). Road traffic forecasting: Recent advances and new challenges. IEEE Intelligent Transportation Systems Magazine, 10(2), 93-109.
4. Nagy, A.M., & Simon, V. (2018). Survey on traffic prediction in smart cities. Pervasive and Mobile Computing, 50, 148-163.
5. Zhang, J., Zheng, Y., & Qi, D. (2017). Deep spatio-temporal residual networks for citywide crowd flows prediction. In Proceedings of the AAAI Conference on Artificial Intelligence (pp. 1655-1661).
6. Ahmed, M.S., & Cook, A.R. (1979). Analysis of freeway traffic time-series data by using Box-Jenkins techniques. Transportation Research Record, 722, 1-9.
7. Williams, B.M., & Hoel, L.A. (2003). Modeling and forecasting vehicular traffic flow as a seasonal ARIMA process: Theoretical basis and empirical results. Journal of Transportation Engineering, 129(6), 664-672.
8. Dougherty, M. (1995). A review of neural networks applied to transport. Transportation Research Part C: Emerging Technologies, 3(4), 247-260.
9. Tian, Y., & Pan, L. (2015). Predicting short-term traffic flow by long short-term memory recurrent neural network. In Proceedings of the IEEE International Conference on Smart City/SocialCom/SustainCom (pp. 153-158).
10. Yu, B., Yin, H., & Zhu, Z. (2018). Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. In Proceedings of the 27th International Joint Conference on Artificial Intelligence (pp. 3634-3640).
11. Zhao, Z., Chen, W., Wu, X., Chen, P.C., & Liu, J. (2017). LSTM network: a deep learning approach for short-term traffic forecast. IET Intelligent Transport Systems, 11(2), 68-75.

12. Friedman, J.H. (2001). Greedy function approximation: a gradient boosting machine. Annals of Statistics, 29(5), 1189-1232.
13. Zhang, Y., & Haghani, A. (2015). A gradient boosting method to improve travel time prediction. Transportation Research Part C: Emerging Technologies, 58, 308-324.
14. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794).
15. Zhang, J., Chen, F., Cui, Z., Guo, Y., & Zhu, Y. (2020). Deep learning architecture for short-term passenger flow forecasting in urban rail transit. IEEE Transactions on Intelligent Transportation Systems.
16. Leshem, G., & Ritov, Y. (2007). Traffic flow prediction using adaboost algorithm with random forests as a weak learner. In Proceedings of the World Academy of Science, Engineering and Technology (pp. 193-198).
17. https://www.kaggle.com/datasets/hasibullahaman/traffic-prediction-dataset